

Drift-Sense Phase 2 “ Failure Analysis

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1. Hand-Rolled FFT-NCC Bug (Root Cause of Phase 1 Accuracy Failures)

What failed: The original `normalized_cross_correlation_fft` (integral image + FFT) had a genuine implementation bug. At the exact ground-truth (scale, theta), it returned wrong locations (18“358px error, score 0.66“0.82) while `cv2.matchTemplate` on the same inputs returned correct locations (0.3“0.7px error, score 0.85“0.87).

Evidence: Direct A/B comparison on real organizer sample pairs at known ground-truth pose. The hand-rolled implementation's score surface had systematic phase errors “ it was not computing normalized cross-correlation correctly despite appearing to produce plausible-looking scores.

Fix: Replaced the core correlation primitive entirely with `cv2.matchTemplate(TM_CC0EFF_NORMED)`. All beam-search and GAR logic retained on top. Result: Set A=1.000, Set B=0.967, Set D=1.000, F1=1.000 vs organizer baseline A=1.000, B=0.467, D=1.000, F1=0.897.

2. Scale-Step Quantization Error

What failed: Stage 1 coarse grid used `scale_step=1.0` (grid: 8,9,10,11,12). For a true scale of e.g. 10.55, the nearest grid point is 11.0 “ a 0.45 mismatch. At this off-scale template, the true embedded position's NCC score degraded enough that an unrelated structural region elsewhere in the search image scored higher by a clear margin. The beam search then refined the wrong branch confidently, producing 81px and 226px localization errors at high confidence (score“0.85, GAR“0.98).

Evidence: At exact true scale (10.55), the true position scored 0.8326 and won outright. At `scale=11.0` (nearest grid), it scored only 0.5621 while an unrelated location scored 0.5875 “ a real, not noise-level gap. Stage 2 theta refinement made it worse: score monotonically increased as theta moved away from true value (-2.3““1.0“), because the wrong location's match landscape was smoother and more favorable at integer scale.

Fix: Changed `scale_step=1.0` “ `scale_step=0.5`. Stage 1 evaluations increased from 25 to 45 (9 scale “ 5 theta anchors). Max scale quantization error reduced from 0.5 to 0.25. Timing impact: ~0.3s additional, negligible vs 17.5s budget. All seeds corrected to sub-1px error.

3. Absent-Pair Generator Design Flaw

What failed: `generate_dataset_v2.py`'s absent-pair branch embedded a same-pitch, same-architecture distractor patch into the search image, while leaving the reference as the true original template. This caused 100% false positive rate on locally-generated Set C: scores 0.62“0.85, all well above rejection threshold.

Root cause: Two renders of the same periodic pattern (same pitch, same architecture, different RNG seed) still correlate strongly under NCC “ `matchTemplate` has 3 degrees of freedom (x, y, and scale/theta already fixed in stage3) to find the best phase alignment between two periodic grids. A sharp, confident false lock (score=0.70, GAR=0.33) resulted, not diffuse aliasing.

Organizer's actual method (confirmed from `phase2_pipeline.py` lines 380“413): search canvas left completely untouched; reference swapped for a decoy from an *independent* canvas with `mat_size=0.55`, `strip_width=2.1` “ deliberately mismatched periodicity that structurally cannot occur anywhere in the true search image. Decoy also force-cropped onto a `mat/strip` junction for additional structural specificity.

Fix: Rewrote absent branch to mirror organizer exactly: `search_clean` left untouched, `ref_clean` replaced with decoy rendered at `pitch=0.55`, `line_width=2.1`, `contact_r=1.6`. Result: false positive rate 20/20“0/20, absent scores now 0.12“0.33, matching real organizer absent range (0.26“0.40).

4. Threshold Calibration “ Borderline Case

What failed: Original thresholds (score“0.42 OR gar“0.60) produced 1 false positive on real organizer data: p018 (absent, Set C severity=0) scored 0.4291 “ just 0.009 above threshold. The score gap between the closest present pair (p012, score=0.4292) and p018 was 0.0001 “ numerically indistinguishable by score alone.

Resolution: GAR provides real separation: p012 GAR=0.6227 (confident, low ambiguity “ true match), p018 GAR=0.7527 (more ambiguous “ no true correspondence). GAR gap = 0.13, a meaningful margin. Changed thresholds to (score“0.43 OR gar“0.65): both p012 and p018 now fail the score branch, but p012 passes the GAR branch (0.6227“0.65) while p018 does not (0.7527>0.65). F1=1.000 with real margin on both axes.

5. Remaining Limitations

Local generator's search background shares pitch/architecture with the reference, causing occasional false locks (1/20) not seen on real data (GAR 0.45“0.86 real vs 0.98+ synthetic) “ all thresholds calibrated

on the real 20-pair set, not local data. Rejection margin is thin with only 4 real absent samples ($p_{018}=0.4291$ vs hardest present $p_{012}=0.4292$); GAR (0.13 gap) resolves it, but the real 200-pair set may include harder decoys per the organizer README, warranting re-check if more absent samples become available. Set D (RGB) handled via grayscale conversion; both pairs localized correctly ($\text{err}<0.32\text{px}$) but no dedicated multi-channel logic implemented.

6. Summary Table

Issue	Impact	Fix	Status
FFT-NCC bug	Wrong location on all pairs	cv2.matchTemplate	Fixed
scale_step=1.0	81â€²226px errors, ~10% pairs	scale_step=0.5	Fixed
Absent-pair generator	100% false positive on local Set C	Mismatched-periodicity decoy	Fixed
Threshold calibration	1 FP on real data (p_{018})	$\text{score}\approx 0.43$, $\text{gar}\approx 0.65$	Fixed
Background self-similarity	Occasional local-data false locks	Generator realism issue; real data unaffected	Documented